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ORIGINAL PAPER

Comparing Virtual Reality, Video, and Audio-Guided Meditations in Fostering Positive Attitudes toward Meditation

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Abstract

Objectives The present study aimed to test the effectiveness of virtual reality (VR) technology to promote positive attitudes toward mindfulness and to enhance motivation to practice mindfulness.

Method Participants ($n = 409$) were randomly assigned to one of four conditions: (1) VR-based meditation, (2) video-based meditation, (3) audio-based meditation, and (4) magazine article reading about meditation as a control group. All three of the meditation conditions featured the same script and audio.

Results Participants in each of the three mindfulness conditions reported an increase in state mindfulness and positive affect, and a decrease in negative affect. Compared to participants who meditated using video or audio clips, participants who meditated using VR reported mindfulness meditation to be more interesting, fun, enjoyable, and less boring. There were no differences in whether participants were distracted, mind wandering, or were absorbed in the meditation across the groups. Participants' self-reported intention to practice mindfulness meditation in the next month and their actual behavior did not differ across the groups.

Conclusions VR meditation could be useful in fostering positive mood and attitudes in regard to mindfulness practice compared to other technology-assisted meditations, but these changes are not necessarily accompanied by stronger intention to meditate in the future.

Keywords Virtual reality (VR) · Mindfulness · Virtual mindfulness · Meditation · State affect

Mindfulness covers a range of contemplative self-regulation techniques, including sitting meditation, moving meditation, and breathing techniques. The core has been defined as a focus on the present moment experience without self-critical judgment (e.g., Kabat-Zinn, 2015). Practicing mindfulness meditation is associated with numerous benefits to mental health, such as feeling of calmness, lower stress, depression, and anxiety symptoms, better cognitive control, and overall improved psychological well-being (Davidson & Kaszniak, 2015; Edenfield & Saeed, 2012; Goyal et al., 2014). Given these benefits, people's interest in mindfulness practice has been growing. Many people, however, often find themselves lost in the process of practicing mindfulness. It is not easy for novice meditators to practice mindfulness on a regular basis, as evidenced by many mindfulness-based intervention studies that report decline in program adherence over time (e.g., Grow et al., 2015). In fact, many people report difficulty in practicing mindfulness as they navigate through their daily lives (e.g., Naumburg, 2016).

One reason for this reported difficulty could be that novice meditators are less capable of concentrating and allocating necessary cognitive resources to practice mindfulness compared to experienced meditators. Those who are unfamiliar with mindfulness may find the meditation experience to be boring, which in turn leads to mind wandering and disengagement with the mindfulness practice (Anderson et al., 2019; Lymeus et al., 2017). Although having a meditation group and qualified instructors could help reduce such problems (e.g., Hanley et al., 2022; Lauricella, 2013), different situational factors (e.g., time, cost, availability) play a role in an individual's ability to find motivation and support for their mindfulness practices.

Mindfulness meditation may be similar to physical exercise in that many people who know it could be good for them are still not easily motivated to do it regularly. The present

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study aims to examine whether the use of virtual reality (VR) technology might be motivational. In recent years, VR has been introduced as a medium to promote mindfulness practices, because VR could provide an immersive, engaging, and controllable environment that reduces distraction, inspires, and promotes relaxation (Fuchs, 2022). Preliminary evidence from smaller-scale studies has shown promising results (e.g., Navarro-Haro et al., 2017; Sevilla-Llewellyn-Jones et al., 2018), though more studies are needed to validate the overall effectiveness of VR in enhancing mindfulness practices. This paper expands on existing literature on VR-based mindfulness practice by focusing on whether VR could change the way people view mindfulness meditation practices, and whether VR could increase behavioral intention to practice mindfulness in the future.

Participation in health behaviors can be predicted by several psychological aspects that come together to influence intention to engage in the behavior (Ajzen & Driver, 1991). According to the theory of planned behavior (Ajzen, 1991), intention to engage in a behavior in the future is influenced by three different determinants: (1) attitudes toward the behavior, (2) subjective norms, which relate to how important others (e.g., family, friends) would approve or disapprove of the action, and (3) perceived behavioral control over the actions (Ajzen & Driver, 1991). Attitude toward behavior is an especially important factor explaining intention when it comes to predicting intention to carry out health behaviors such as physical exercise (Ajzen & Manstead, 2016; Godin & Kok, 1996). The stronger the intention is, the more motivated an individual is to perform a behavior (Ajzen, 1991). Thus, it is expected that people's intention to practice mindfulness would be stronger if they hold more favorable attitudes toward mindfulness meditation. This is not to say that increasing the normative beliefs of social support and perceived self-efficacy would not also increase intentions, just that changing attitudes might be easier. To increase the intention to meditate, therefore, we could focus on fostering positive attitudes towards mindfulness meditation with a positive experience. It is worth noting that attitudes are not unitary constructs, but have emotional, cognitive, and experiential aspects (Ostrom, 1969). It would therefore be useful to measure emotional responses, cognitive aspects such as absorption and distractibility, and experiential evaluations of a mindfulness practice.

VR technology might elicit stronger motivation to meditate by making mindfulness practice a more engaging experience. A VR headset provides slightly different pictures to each eye, allowing for the perception of a 3D environment (Mukamal, 2017). By wearing a headset and headphones, one can perceive oneself to be "in" an environment. Simultaneous synchronous visual and auditory components combine to simulate a real environment as people's physical movements (e.g., turning the head) is associated with relevant changes in the virtual environment (e.g., change of the scene, sound of the footsteps). Such illusion allows participants to experience the feeling of "being there" (Cebolla et al., 2019; Cooper et al., 2018; Slater, 2009). This may be particularly useful for novice meditators, as they immerse into the environment with little or no distraction from the real world (Costa et al., 2019). VR further allows practitioners to experience higher degree of presence (Bennett, 2018; Riva et al., 2011) and decentering (Chandrasiri et al., 2019), along with lower distraction (Waller et al., 2021). As a result, VR-based mindfulness meditation can lead to a significant increase in state mindfulness (Chandrasiri et al., 2019; Seabrook et al., 2020; Yildirim & O'Grady, 2020) just as audio-based and video-based mindfulness meditation can (Bravo et al., 2018; Johnson et al., 2015; Waller et al., 2021; Yildirim & O'Grady, 2020).

High immersion and engagement through VR mindfulness meditation have been associated with more positive attitudes and evaluations of mindfulness practice (e.g., Kolbe et al., 2021; Soh et al., 2021). For example, participants who practiced mindfulness through VR perceived relaxation and meditation techniques to be more beneficial compared to those who practice mindfulness through non-interactive videos (Kaplan-Rakowski et al., 2021). Participants who practiced mindfulness using VR also reported more positive moods (Botella et al., 2015; Miller et al., 2021; Navarro-Haro et al., 2017; Seabrook et al., 2020; Tarrant et al., 2018) as well as less negative affect, such as anxiety and depression (Chavez et al., 2020; Flores et al., 2018; Ge & Han, 2021; Wren et al., 2021). The decrease in negative affect was also supported by neurological evidence that participants who experienced guided VR meditation intervention (*vs.* no-meditation control) showed a global decrease in Beta activity, indicating a state of decreased anxiety and hyperarousal (Tarrant et al., 2018). In the study by Mistry and others (2020), participants who practiced mindfulness using VR (*vs.* non-VR mindfulness group) reported feeling more awe, wonder, and amazement. They also reported higher level of serenity, content, and peacefulness, in addition to lower bodily discomfort and mind wandering. In sum, VR-based mindfulness meditation could offer a more positive meditation experience to the practitioners. It is, however, yet unclear whether these positive changes in affect are unique or stronger for VR-based meditation, with some studies suggesting that VR-based meditation can elicit more positive affect changes compared to non-VR meditations (e.g., Tarrant et al., 2022), and some studies suggesting no significant difference between VR and non-VR meditations in state affect (e.g., Waller et al., 2021).

Furthermore, VR technology is also an ideal candidate for enhancing perceived behavioral control, which is another determinant of intention according to the theory of planned behavior (Ajzen, 1991). Beyond having a motivation to execute a behavior, one must also believe that one has sufficient

self-efficacy to be able to do it. VR technology can give people the ability to actively interact and practice behaviors that might not be as possible in their "real" lives. VR technology allows people to feel a sense of competence and autonomy, as they can move their body freely to control their environment, which is not possible with stationary non-interactive videos (De Gauquier et al., 2019; Lau & Lee, 2019).

VR technology is already in use for different psychological treatments and behavioral changes, such as smoking cessation and pain management, among others (Goldenhersch et al., 2020; Gómez-Busto & Ortiz, 2020; Haisley et al., 2020; Lafortune et al., 2020; Navarro-Haro et al., 2019; Qian et al., 2020; Wren et al., 2021), with the rationale that intense level of engagement and full immersion in the virtual environment could lead to long-term adherence to the desired behavior (e.g., Mestre et al., 2011). Similar principles could be applied to mindfulness practices to promote practitioners' perceived behavioral control (Navarro-Haro et al., 2017; Sliwinski et al., 2017).

More studies, however, are needed to validate the effectiveness of VR technology in increasing mindfulness practice. Not all intervention studies had an alternative meditation group when evaluating the use of VR mindfulness practice (e.g., Crosswell & Yun, 2020; Keijsers et al., 2021; Weitzman et al., 2021; Wren et al., 2021). Similarly, many past experimental studies on VR-based mindfulness did not include an active control group. A recent review by Arpaia and others (2021) found that only 27% of the studies had a control group, whereas another review by Zhang and others (2021) found that five out of nine identified studies had comparable baseline measures for both the treatment and the control group, suggesting possible room for methodological improvements. In addition, many preliminary studies had a small sample size per cell (e.g., Chandrasiri et al., 2019; Crosswell & Yun, 2020; Kolbe et al., 2021; Navarro-Haro et al., 2017; Seabrook et al., 2020; Weitzman et al., 2021). A more statistically-powered experimental study with an active control group and alternative comparison groups would be beneficial in validating the effectiveness of VR technology in fostering positive attitudes towards mindfulness meditation and enhancing intention to practice mindfulness.

The aim of the present study was to use a large sample of young adults in the U.S. to compare whether a VR-based mindfulness meditation exercise increased attitudes and intentions relative to other technology-assisted meditation practices (e.g., video-based meditation, audio-based meditation) and a control group. It was expected that participants' state mindfulness would increase after practicing mindfulness in all meditation conditions, and the increase would be the greatest for participants who used VR (H1). Positive affect should increase and negative affect should decrease after the meditation, and the changes in affect would also be the greatest for the VR mindfulness meditation condition (H2). We also hypothesized that participants in the VR meditation condition would have a positive experience during their meditation session, such as more absorption, reduced mind wandering, and lower distraction (H3), have more positive attitudes toward future mindfulness meditation (H4), and stronger intention to practice mindfulness in the future (H5) compared to the participants in other conditions.

Method

Participants

Undergraduate students ($n = 409$) from a large Midwestern university in the United States took part in the experiment in exchange for course credit. Participants were eligible for the study if they were at least 18 years old and did not have a history of photosensitive seizures. Participants were majority female (60.9%) with an average age of 19.4 ($SD_{age} = 1.73$), and 89.6% of the participants spoke English as their first language. Most participants self-identified as White (74.2%), followed by Asian or Pacific Islander (8.5%), Latinx or Hispanic (5.6%), and African Americans (4.3%); the remaining participants reported being multi-racial or other racial category. Most participants did not have any prior meditation experience (60.1%) or had prior experience but do not meditate on a regular basis (27.2%). A small percentage of the participants (12.7%) said they meditated regularly (i.e., at least once a week). Post-hoc power analysis conducted using G*Power version 3.1 (Faul et al., 2007) indicated that the sample size used in this study was sufficient to achieve a power of 0.95, assuming a small effect size ($d = 0.15$).

Procedure

After the informed consent procedure, participants completed measures on state mindfulness and state affect. They were randomly assigned to one of the four conditions: (1) Virtual reality meditation ($n = 102$), (2) Video meditation ($n = 103$), (3) Audio meditation ($n = 101$), or (4) Meditation article reading ($n = 103$). All procedures were administered by trained research assistants.

In the virtual reality (VR) meditation condition, participants experienced two guided meditations from the video game *Guided Meditation VR* using the HTC Vive headset and headphones (https://guidedmeditationvr.com/).[1] Two specific meditations were chosen from the program. The first

[1] The publisher of the game did not specify whether the meditations are from evidence-based practices. The scripts seem to us, however, to be very similar to typical beginning mindfulness and self-compassion guided meditations.

Fig. 1 Screenshots of the virtual environments

meditation was set in a waterfall with a male voice guiding a basic relaxing mindfulness of breath meditation (https://guidedmeditationvr.com/zen-meditation-2/), and the second meditation was in an autumn glade with a female voice guiding a self-compassion meditation. Both meditations had light background music and natural sound effects from the scenes (e.g., water flow, rustling of leaves). Participants could turn their heads to look at the virtual scenery around them while meditating (see Fig. 1 for screenshots of the virtual environment). Each meditation was about 10 min long. The total duration of the sessions was chosen partly based on previous findings that participants who are not familiar with the VR technology had harder time maintaining their attention if the duration is longer than 20 min (Mazgelytė et al., 2021), and also to allow all measures to fit within a 60-min study.

The video meditation condition used the identical audio from the VR meditation condition but were paired with videos of a water fall and autumn glade, respectively. These were films of real nature scenes, not virtual. The camera was stationary, and the scene could not be explored as in the VR condition.

The audio meditation condition used the same audio as with the VR and video meditation conditions, but there was no visual component. Examples comparing the three meditation conditions for the two practices can be found in the supplementary material.

A meditation article condition was an active control condition where the participants did not practice meditation. Instead, participants read an 8-page *Mindful* magazine article titled "The science of meditation" that described the benefits of meditation (Delehanty, 2017). This was chosen (1) to give participants something to do that would also take about 20 min, and (2) might possibly also provide some motivation to meditate in the future, although it should not have any short-term effects on state mindfulness or emotions. A non-active control condition, such as sitting and waiting for 20 min, was not chosen for this study for several reasons. First, the post-activity questionnaire would not make sense for someone who had not been primed to consider meditation. Second, it is becoming more standard practice in intervention studies to have an active control, both because it helps to blind the participant to what condition they are in and also because it can be argued to be more ethical when a benefit is expected to occur from the intervention (e.g., Saks et al., 2002).

After either completing the meditation practice or reading the article, participants completed post-intervention state measures of mindfulness and affect. They also answered a set of questionnaires on their meditation experience, meditation intentions and attitudes, willingness to meditate, as well as demographic information. Finally, a behavioral measure of participants' intention to engage in mindfulness meditation in the future was measured. Participants were debriefed and thanked.

Measures

State mindfulness

State mindfulness was measured using the Toronto Mindfulness Scale (TMS; Lau et al., 2006), which is a 5-point Likert-type scale (1 = not at all, 5 = very much) that assesses state-like experience of meditation. TMS consists of two subscales: Decentering (7 items) and curiosity (6 items). The sample items include "I experienced myself as separate from my changing thoughts and feelings" (decentering subscale) and "I was curious to see what my mind was up to from moment to moment" (curiosity subscale). For this sample, the Cronbach's alphas for the curiosity subscale were 0.86 (pre-intervention) and 0.93 (post-intervention), decentering = 0.66 (pre) and 0.82 (post), and for the entire scale = 0.80 (pre) and 0.91 (post). McDonald's omegas for the curiosity subscale were 0.66 (pre-intervention) and 0.82 (post-intervention), decentering = 0.87 (pre) and 0.93 (post), and for the entire scale = 0.83 (pre) and 0.91 (post).

State affect

Current positive and negative emotions were measured by asking the participants to rate the extent to which they felt each of 12 emotions in the past 15 min on a 5-point Likert-type scale (1 = *not at all*, 5 = *extremely*). Six positive (e.g., happy, relaxed, energized, caring, peaceful, curious) and six negative emotions (e.g., sad, anger, anxious, bored, stressed, surprise) were measured. Surprise did not load with other items on the scale and was thus dropped from the final analysis. The Cronbach's alphas for the pre- and post-intervention measures of positive emotions were 0.73 and 0.87, and those for the negative emotions were 0.71 and 0.66, respectively. McDonald's omegas for the pre- and post-intervention measures of positive emotions were 0.78 and 0.86, and those for the negative emotions were 0.74 and 0.63, respectively.

Experiential Evaluation

Participants who were assigned to one of the three meditation conditions (e.g., VR meditation, video meditation, and audio meditation) evaluated their experience on a 7-point Likert-type scale (1 = *strongly disagree*, 7 = *strongly agree*) using the following three statements: (1) I thought of other things during the meditation (distraction), (2) I was entirely absorbed in the meditation (absorption), and (3) I found my mind wandering while I was supposed to be meditating (mind wandering).

Level of Engagement

Participants in the meditation conditions rated to what extent they agree that the guided meditation was "very interesting" (interest), "a boring activity" (boredom), "enjoyable" (enjoyment), "a waste of time", and "fun" on a 5-point Likert scale, with 1 = *disagree*, 5 = *agree*.

Acceptability of Intervention

Participants who meditated were asked to report the perceived satisfaction, usefulness, and difficulty to use related to the meditation program. Participants also indicated how likely they would be to use the given type of meditation tool if they had easy access to it. The response range varied from 0 (*not at all*) to 10 (*very much*).

Intention to Meditate

A single item was used to measure participant's self-reported willingness to engage in meditation, using the following question: "How many minutes of meditation do you plan to engage per day for the following week?" Participants answered the question in number of minutes per day. We specifically asked participants to report expected time meditating the following week to capture participants' intention to meditate with their social circumstances taken into consideration, as intention is one important factor in predicting behavior (e.g., Gibbons et al., 1998a, b; Gibbons et al., 1998a, b).

Meditation Intentions and Attitudes

Ten items were developed for this study to capture participants' intention to meditate in the future and positive attitudes toward future meditation. Measures included intention to meditate, positive attitudes toward mindfulness, subjective norms, and perceived behavioral control. Participants read the statements such as "I intend to meditate most days, if not all days of the week, for the next month" (intentions) and "For me to meditate most days, if not all days of the week, for the next month would be beneficial" (positive attitudes), then rated how much they agreed to each statement on a 5-point Likert scale (1 = *strongly disagree*, 5 = *strongly agree*).

Behavioral Measure of Intention to Meditate in the Future

At the end of the study, participants were given a chance to take a free copy of information on mindfulness meditation groups that meet near campus, which served as a proxy measure of intention to practice mindfulness in the future. Participants' behavior was coded as a binary outcome, such that 0 = did not take the brochure (e.g., no intention to engage in mindfulness meditation) and 1 = took the brochure (e.g., has intention to engage in mindfulness meditation). Given that it might be easy to claim an intention to meditate, this was included to provide a more stringent test of interest/intention to meditate.

Demographics

Participants reported their age, sex, race, and their prior meditation experience. They also indicated whether English was their first language.

Data Analyses

Analyses were conducted using IBM SPSS version 22. Descriptive statistics were computed for state mindfulness (pre- and post-intervention), state affect (pre- and post-intervention), experiential evaluation, attitudes toward meditation, and future meditation intention. Repeated measures ANOVAs with planned contrasts were used to compare the changes in state mindfulness, state positive affect, and state negative affect following the intervention among the four groups: (1) Virtual reality meditation, (2) Video meditation,

Table 1 Mean and standard deviation (in parentheses) for state mindfulness (pre- and post-interventions), with mean differences and standard error (in parentheses)

		1. VR meditation ($n=100$)	2. Video meditation ($n=102$)	3. Audio meditation ($n=101$)	4. Article reading ($n=103$)
State mindfulness	Pre	3.23 (0.60)	3.21 (0.54)	3.27 (0.56)	3.21 (0.62)
	Post	3.46 (0.68)	3.37 (0.69)	3.28 (0.75)	3.19 (0.72)
Mean difference	1	-			
	2	0.05 (0.08)	-		
	3	0.07 (0.08)	0.02 (0.08)	-	
	4	0.15 (0.08)	0.09 (0.08)	0.07 (0.08)	-

Table 2 Mean and standard deviations (in parentheses) for state affect measures (pre- and post-interventions), with mean differences and standard error (in parentheses)

		1. VR ($n=100$)	2. Video ($n=102$)	3. Audio ($n=101$)	4. Article reading ($n=103$)
Positive emotion	Pre-test	3.06 (0.75)	2.96 (0.82)	2.84 (0.84)	2.95 (0.85)
	Post-test	3.33 (0.82)	3.35 (0.92)	3.07 (0.90)	2.54 (0.84)
Mean difference	1	-			
	2	0.04 (0.11)	-		
	3	0.24 (0.11)*	0.20 (0.11)	-	
	4	0.45 (0.11)***	0.41 (0.11)***	0.21 (0.11)	-
Negative emotion	Pre-test	2.03 (0.71)	2.18 (0.76)	2.10 (0.68)	2.00 (0.83)
	Post-test	1.53 (0.55)	1.62 (0.58)	1.76 (0.70)	1.75 (0.60)
Mean difference	1	-			
	2	-0.12 (0.08)	-		
	3	-0.15 (0.08)	-0.03 (0.08)	-	
	4	-0.09 (0.08)	0.03 (0.08)	0.05 (0.08)	-

* $p<0.05$, *** $p<0.001$

(3) Audio meditation, and (4) Meditation article reading (control group). One-way ANOVA with Bonferroni corrected post-hoc tests were conducted to compare participants' experiential evaluation among the three meditation groups: (1) Virtual reality meditation, (2) Video meditation, and (3) Audio meditation. Similarly, one-way ANOVA with Bonferroni corrected post-hoc tests were conducted to compare participants' attitudes toward meditation and future meditation intention among all four groups. Effect sizes (partial η^2) were also reported in all ANOVAs. Finally, a Chi-square goodness of fit test was carried out to compare the number of participants in each group who took the meditation group brochure.

Results

Participants' sex and prior meditation experience were included as covariates in all analyses where appropriate to control for their possible effect on the dependent variables. Table 1 summarizes the descriptive statistics for state mindfulness, measured before and after the interventions. Participants who did not provide complete measures on state mindfulness and state affect were removed from the analysis ($n=3$).

A repeated measures ANOVA was used to test whether participants' state mindfulness differed by group (H1). There was a significant main effect of time points on participants' state mindfulness measures ($F(1, 402)=13.61$, $p<0.001$, $\eta^2=0.03$). Post-hoc analyses revealed that participants in the VR meditation condition and video meditation condition had an increase in their state mindfulness scores (p-values <0.05). No significant increase in state mindfulness was observed for participants who participated in audio meditation or read the article on meditation. Contrary to the initial hypothesis that the increase in state mindfulness would be the greatest for the VR meditation condition, no significant mean difference was observed when comparing VR meditation condition and video meditation condition (mean difference $=0.05$ [95% *CI*: -0.11 to 0.22], $p=0.50$).

Participants' state affect scores (pre- and post-intervention) were compared to test the hypothesis that mindfulness practice will change positive and negative affect (H2). Table 2 summarizes results comparing state affect scores.

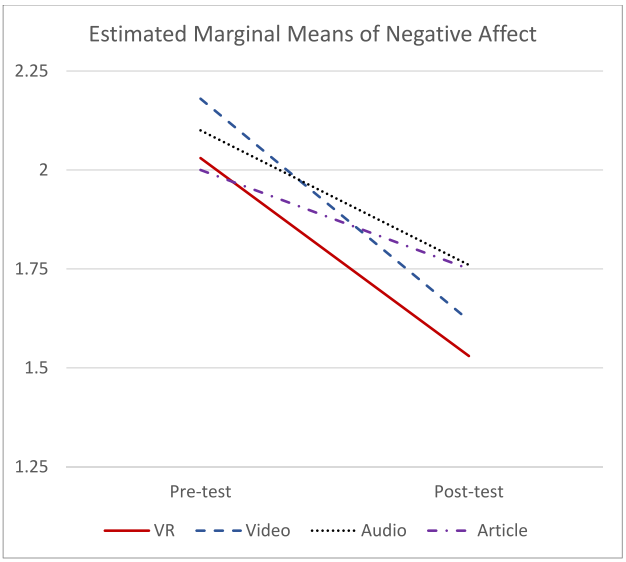

Fig. 2 Changes in Negative Emotion by Condition

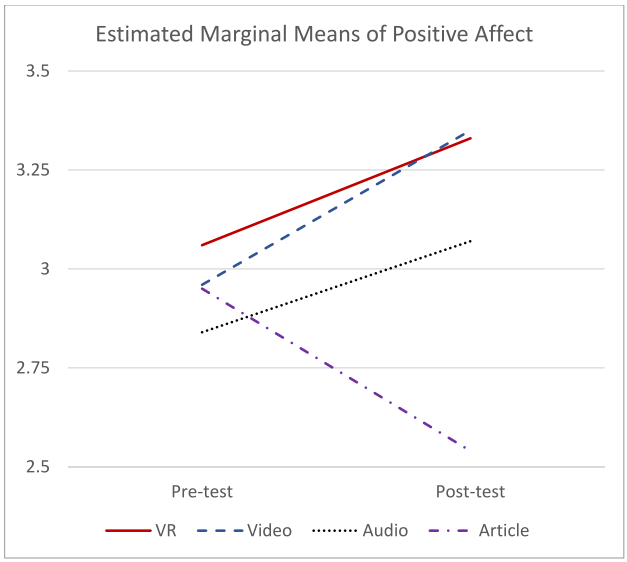

Fig. 3 Changes in Positive Emotion by Condition

There were no significant mean differences in baseline (pre-intervention) measures of positive or negative affect across four groups; that is, at pre-test all four groups were similar. A repeated measures ANOVA indicated that there was a significant pre-post change in negative state affect, $F(1, 402) = 159.63$, $p < 0.001$, $\eta^2 = 0.08$. Participants in all four conditions reported decreased negative affect, with no significant differences among the four conditions (Fig. 2).

Regarding positive state emotion, there was also an overall significant mean difference in pre- and post-intervention positive state affect, $F(1, 402) = 12.50$, $p < 0.001$, $\eta^2 = 0.01$, and the direction of change differed among the groups. Participants in the article reading condition reported decreased positive affect after the intervention, whereas participants in the other three conditions (e.g., VR meditation, video meditation, audio meditation) showed increased positive affect (Fig. 3).

Post-hoc comparisons revealed that participants in the VR meditation condition reported greater increase in positive affect compared to those in the audio meditation condition (mean difference = 0.24 [95% *CI*: 0.03 to 0.45], $p = 0.03$) and the article reading condition (mean difference = 0.45 [95% *CI*: 0.24 to 0.66], $p < 0.001$). Similarly, participants in the video meditation condition had a greater increase in positive affect compared to those in the article reading condition (mean difference = 0.41 [95% *CI*: 0.20 to 0.62], $p < 0.001$).

One-way ANOVA with Bonferroni corrected post-hoc tests were used to analyze participants' evaluations of the interventions (e.g., experiential evaluation). These data were only collected from the participants who took part in a meditation practice (i.e., excluding the article reading control group). Table 3 summarizes the descriptive and inferential statistics. There were no significant mean differences in measures of experiential evaluation (e.g., distraction, absorption, mind wandering) between the three meditation groups.

There were no mean differences in whether participants perceived the meditation practice as a waste of time. There were, however, significant mean differences for all other measures of meditation engagement. Post-hoc tests revealed that VR meditation group perceived the meditation as more interesting (higher interest score) compared to the video meditation (mean difference = 0.38 [95% *CI*: 0.01 to 0.75], $p = 0.05$) and audio meditation group (mean difference = 0.61 [95% *CI*: 0.13 to 0.89], $p = 0.004$). The VR meditation group also perceived the meditation as less boring (lower boredom score) and more enjoyable (higher enjoyment score) than the audio meditation group only (mean difference = -0.44 [95% *CI*: -0.83 to -0.06], $p = 0.018$ and mean difference = 0.40 [95% *CI*: 0.06 to 0.74], $p = 0.014$, respectively). Finally, the VR meditation group rated the meditation practice as more fun compared to the video meditation (mean difference = 0.37 [95% *CI*: 0.02 to 0.73], $p = 0.038$) and audio meditation groups (mean difference = 0.43 [95% *CI*: 0.07 to 0.79], $p = 0.013$). There were no significant mean differences in ratings between the video meditation and audio meditation groups.

For the acceptability of meditation, those in the VR meditation group reported that they were more likely to use the meditation program if they had easy access to it than the video meditation group did (mean difference = 1.03 [95% *CI*: 0.01 to 2.06], $p = 0.046$). There were no mean differences for participants' rating of satisfaction, perceived usefulness, and difficulty to use the meditation program based on the

Table 3 Summary of descriptive and inferential statistics for experiential evaluation measures

	M (*SD*)	F	MSE	η^2
Experiential evaluation: Distraction		0.40	0.92	0.01
VR meditation	4.72 (1.51)			
Video mediation	4.77 (1.61)			
Audio meditation	4.89 (1.57)			
Experiential evaluation: Absorption		0.37	0.88	0.01
VR meditation	4.47 (1.48)			
Video mediation	4.40 (1.59)			
Audio meditation	4.19 (1.60)			
Experiential evaluation: Mind wandering		1.56	3.26	0.01
VR meditation	4.61 (1.54)			
Video mediation	4.58 (1.57)			
Audio meditation	4.93 (1.33)			
Level of engagement: Interest		5.70**	6.64	0.04
VR meditation	4.19 (0.92)			
Video mediation	3.76 (1.21)			
Audio meditation	3.61 (1.21)			
Level of engagement: Boredom		4.41*	5.33	0.03
VR meditation	2.01 (1.08)			
Video mediation	2.34 (1.18)			
Audio meditation	2.43 (1.19)			
Level of engagement: Enjoyment		4.11*	5.33	0.03
VR meditation	4.27 (0.76)			
Video mediation	4.01 (1.04)			
Audio meditation	3.84 (1.08)			
Level of engagement: Waste of time		3.00	2.69	0.02
VR meditation	1.69 (1.03)			
Video mediation	1.91 (1.07)			
Audio meditation	1.89 (1.06)			
Level of engagement: Fun		4.90**	5.21	0.03
VR meditation	3.71 (1.08)			
Video mediation	3.30 (1.08)			
Audio meditation	3.21 (1.00)			
Acceptability of intervention: Satisfaction		2.43	12.14	0.02
VR meditation	8.00 (2.01)			
Video mediation	7.50 (2.47)			
Audio meditation	7.18 (2.31)			
Acceptability of intervention: Usefulness		0.68	4.14	0.01
VR meditation	8.16 (2.30)			
Video mediation	7.72 (2.84)			
Audio meditation	7.66 (2.38)			
Acceptability of intervention: Difficulty to use		0.55	3.32	0.01
VR meditation	3.66 (2.71)			
Video mediation	3.71 (2.49)			
Audio meditation	3.79 (2.57)			
Acceptability of intervention: Likely to use in the future		3.59*	31.18	0.03
VR meditation	7.82 (2.70)			
Video mediation	6.57 (3.20)			
Audio meditation	6.84 (3.00)			
Willingness to meditate (in minutes)		2.25	306.69	0.01
VR meditation	13.28 (12.64)			
Video mediation	12.43 (12.42)			

Table 3 (continued)

	M (*SD*)	F	MSE	η^2
Audio meditation	10.02 (9.78)			

* $p < 0.05$; ** $p < 0.01$

condition. Finally, the mean minutes participants intended to spend per day to meditate in the following week did not differ by meditation intervention condition.

Participants' attitudes toward meditation and future meditation intention were compared across the four intervention groups. There were significant mean differences among the groups on whether they believed important social others think they should meditate more regularly. Post-hoc tests with Bonferroni correction demonstrated that participants in the video condition were more likely to believe that important people around them would think they should meditate regularly for the next month compared to the article reading control group (mean difference = 0.47 [95% *CI*: 0.05 to 0.89], $p = 0.018$). It is unclear, however, why perceived social support would differ by condition. The groups did not significantly differ in other attitudes and intention measures. Table 4 summarizes the results.

Finally, a Chi-square goodness of fit test was used to determine whether the number of participants who took a mindfulness meditation group information brochure (e.g., proxy measure of behavioral intention to meditate) differed by condition. Table 5 shows the contingency table. The four groups did not differ in the number of participants taking the information brochure, $\chi^2(1, 3) = 4.44$, $p = 0.22$ (although only 23 out of 409 people took it).

Discussion

The goal of the study was to evaluate the effectiveness of VR mindfulness meditation on promoting positive attitudes towards mindfulness practice and future intention to meditate compared to identical content in video-based, audio-based meditations, and a control group, using a large general undergraduate student sample. Specifically, the present study focused on participants' evaluation of the meditation exercise and intention to practice meditation in the future, as well as changes to participants' state mindfulness and state affect. The study has the strength that it compared three different types of technology-assisted mindfulness meditations with an active control group to rule out the possibility that the effects observed were solely due to the mindfulness practice. The findings of the current study conceptually replicated past research studies by demonstrating that VR meditation increased participants' state mindfulness (Seabrook et al., 2020; Yildirim & O'Grady, 2020). In fact, both VR meditation and video-based meditation led to an increase in state mindfulness, whereas the audio-based meditation and article reading control condition did not. This is inconsistent with the initial hypothesis (H1), where we hypothesized that participants' state mindfulness should increase after practicing mindfulness, regardless of the medium. Participants who used audio-guided meditation reported lower levels of absorption and higher levels of distraction and mind wandering compared to the VR meditation and video meditation groups; the differences between VR meditation and video meditation groups were not statistically significant. Such results may demonstrate that certain mediums of mindfulness meditation practice may be more effective in inducing increased state mindfulness. Prior studies have focused more on differences in content among meditation apps, rather than on differences in delivery with the same content (e.g., Roquet & Sas, 2018).

In alignment with H2, all three meditation groups increased positive affect, successfully replicating the finding that VR meditation induced more positive mood (Botella et al., 2015; Miller et al., 2021; Navarro-Haro et al., 2017; Tarrant et al., 2018). In addition, participants in the VR and video meditation groups had greater increases in positive affect compared to the audio meditation group. There were, however, no significant differences between the increase in positive affect for VR and video meditation groups.

The finding that VR meditation was effective in reducing negative affect (Chavez et al., 2020; Flores et al., 2018; Ge & Han, 2021; Wren et al., 2021) was also replicated. Participants in all intervention conditions (e.g., VR, video, audio, article reading) reported a decrease in their negative affect, and the degree of reduction in negative affect did not differ across the four conditions. It is interesting to note that participants in the control group (e.g., reading a magazine article describing the benefits of meditation) also experienced a decrease in negative affect. Although it is beyond the scope of the present study, it would be beneficial to examine whether describing the benefits of meditation itself could act as a cognitive prime to reduce anxiety and distress (this condition also found lowered positive emotion, however, suggesting that perhaps reading simply lowered active emotional states in general).

We initially hypothesized that participants in VR meditation condition would experience more positive experiences from mindfulness meditation exercises (H3). Contrary to the hypothesis, no significant differences were observed for any self-reported measures of experiential evaluation: Participants in VR, video, and audio meditation groups did not

Table 4 Summary of descriptive and inferential statistics for attitudes toward meditation and future meditation intention

	M (*SD*)	F	MSE	η^2
I intend to meditate most days, if not all days of the week, for the next month		0.33	0.47	0.01
VR meditation	2.67 (1.19)			
Video mediation	2.65 (1.34)			
Audio meditation	2.67 (1.17)			
Article reading	2.82 (1.26)			
I will try to meditate regularly this next month		0.66	0.98	0.01
VR meditation	3.21 (1.17)			
Video mediation	3.30 (1.35)			
Audio meditation	3.09 (1.23)			
Article reading	3.10 (1.26)			
Most of the people who are important in my life would think I should meditate regularly for the next month		3.81*	4.73	0.03
VR meditation	3.00 (1.12)			
Video mediation	3.06 (1.20)			
Audio meditation	2.72 (1.10)			
Article reading	2.61 (1.09)			
I would receive a great deal of support from the people closest to me in my efforts to meditate regularly this next month		1.07	1.16	0.01
VR meditation	3.45 (0.99)			
Video mediation	3.42 (1.13)			
Audio meditation	3.20 (1.01)			
Article reading	3.34 (1.01)			
All things considered, if I wanted to I could easily meditate most days, if not all days of the week, this next month		0.74	0.82	0.01
VR meditation	3.45 (1.01)			
Video mediation	3.61 (1.05)			
Audio meditation	3.45 (1.10)			
Article reading	3.47 (1.11)			
I have very little power over my ability to meditate regularly this next month		0.18	0.14	0.01
VR meditation	2.23 (0.90)			
Video mediation	2.17 (0.92)			
Audio meditation	2.19 (0.78)			
Article reading	2.14 (1.02)			
For me to meditate most days, if not all days of the week, for the next month would be valuable		0.64	0.60	0.01
VR meditation	3.74 (0.80)			
Video mediation	3.57 (1.13)			
Audio meditation	3.54 (0.97)			
Article reading	3.62 (0.99)			
For me to meditate most days, if not all days of the week, for the next month would be pleasant		2.60	2.20	0.02
VR meditation	3.92 (0.64)			
Video mediation	3.63 (1.06)			
Audio meditation	3.60 (0.99)			
Article reading	3.59 (0.96)			
For me to meditate most days, if not all days of the week, for the next month would be beneficial		0.92	0.71	0.01
VR meditation	3.97 (0.75)			
Video mediation	3.75 (1.03)			
Audio meditation	3.86 (0.85)			
Article reading	3.82 (0.91)			
For me to meditate most days, if not all days of the week, for the next month would be fun		2.04	1.72	0.02
VR meditation	3.46 (0.81)			
Video mediation	3.15 (0.94)			

Table 4 (continued)

	M (*SD*)	F	MSE	η^2
Audio meditation	3.21 (0.94)			
Article reading	3.19 (0.93)			

* $p < 0.05$

Table 5 Number of participants who took the meditation brochure, by condition

	VR	Video	Audio	Article reading	Total
Took the brochure	4	5	4	10	23
Did not take the brochure	98	98	97	93	386
Total	102	103	101	103	409

differ in their level of distraction, absorption, and mind wandering ratings. These results of the current study are not consistent with past findings (e.g., Bennett, 2018; Mistry et al., 2020; Mrazek et al., 2012; Riva et al., 2011). It is possible that a brief 20-min mindfulness practice is not sufficient to bring significant changes to participants' feelings of absorption, distraction, and mind wandering. Indeed, some previous studies suggest that VR may not result in better attention to the environment (e.g., Barton et al., 2020; Yildirim & O'Grady, 2020). A more parsimonious explanation, however, may be that technology really doesn't matter; the practice works equally well (in terms of absorption, distraction, and mind wandering) regardless of delivery mechanisms.

Nonetheless, VR is a relatively novel technology that not everyone is familiar with. Individual differences, such as trait absorption, can yield different experiences to VR meditation (e.g., Miller et al., 2021). Future studies could focus on whether such individual differences play a role in practitioners' differing experiences in the VR environments.

There were mixed results regarding participants' attitudes toward the meditation across the three intervention groups. H4 was partially supported in that participants in the VR meditation group reported the most favorable attitudes toward meditation. Compared to video and audio meditation groups, VR meditation group perceived the mindfulness exercise as more interesting, enjoyable, fun, and less boring. The VR meditation group were more likely to report that they would like to use the meditation program if given easy access to it. In addition, VR meditation group and other meditation groups did not differ in perceived difficulty of the meditation program. It is important to note that participants from all three groups reported lower scores on whether they thought the meditation exercise was a waste of time, suggesting that mindfulness meditation, regardless of its delivery method, was perceived as a valuable use of the time to the participants. Yet the positive attitudes toward the current meditation did not translate into greater intention to meditate in the future, as indicated by the fact that there were no group differences in their ratings on how valuable, pleasant, beneficial, or fun it would be to meditate for the next month. The three groups also did not differ in the level of satisfaction and perceived usefulness of the meditation program. Although these results were inconsistent with our hypothesis, the results nonetheless are deemed promising because it means that video and audio-based guided meditation tools, which are more accessible and affordable, could be as satisfying and useful as VR-based meditation.

Finally, participants' intention to practice mindfulness in the future did not differ by their intervention group. Thus, we did not find sufficient evidence to support H5. For example, participants did not differ in the amount of time they are willing to devote per day in the future to practice mindfulness. Participants also did not differ in their perceived behavioral control in practicing mindfulness in the next month. Those in the video meditation group (*vs.* article reading group) were more likely to hold the normative belief that people around them would support them practicing mindfulness, although the effect was relatively small. Finally, contrary to the hypothesis, participants who used VR meditation did not take more information brochures to learn more about attending a local mindfulness meditation group. That is, the brief mindfulness meditation did not result in immediate changes in intention to practice mindfulness in the near future, despite the positive changes to state mindfulness, state affect, and experiential evaluations. In other words, although participants who meditated using VR technology perceived VR meditation more positively, such favorable view was not sufficient to induce behavioral changes. It would be helpful to examine whether multiple sessions of mindfulness meditation in the long run would result in changes in participants' positive attitudes toward mindfulness practice and intention to practice mindfulness.

Another interpretation is that a single 20-min session may be sufficient to raise awareness of meditation, but is not sufficient to increase intentions. Within the theory of planned behavior, attitudes, subjective norms, and perceived ability to do the behavior all work together to influence intentions. The VR condition did have more positive attitudes about meditation. The video condition had higher beliefs that significant others would support them practicing mindful

meditation. All conditions seemed equal in difficulty of use, which is an aspect of perceived behavioral control. The VR condition reported greater willingness to use it if they had access to VR technology, but since most people do not (Statista, 2023), this reduces perceived behavioral control. It seems clear, therefore, that no condition was sufficiently better than the others in enhancing intention because none improved attitudes, subjective norms, or perceived behavioral control sufficiently to increase intention significantly, at least within this single session with a naïve undergraduate sample.

Limitations and Future Directions

There are a few limitations that should be noted when interpreting the findings of the study. First, participants were drawn from a pool of undergraduate students, who are mostly in their late teens and early 20s. In this study, participants in the VR meditation condition did not report significant differences in difficulty to use VR. People from different age groups may react differently to VR technology, particularly if they are not familiar with it. Similarly, the majority were White (which is representative of the university population). Future studies should consider whether the findings of this study can be generalized to different age groups. Likewise, given that most participants were White, it would be beneficial to further explore whether the finding can also be generalized to different racial and ethnic groups.

Similarly, other characteristics of the participants should be considered when studying the effect of VR meditation. Only a small portion of the participants (12.7%) of the present study reported that they meditate at least once a week. The meditation experience is expected to differ based on the practitioner's prior knowledge and practices with regard to meditation. Therefore, future research could examine whether novice *vs.* expert meditators show different results to mindfulness meditation delivered through different technologies. In similar vein, other individual differences such as trait absorption and susceptibility to narrative transportation could be considered.

It is important to note that the result of the study cannot be used to infer the usefulness of VR in mindfulness-based interventions for several reasons. First, this was a single 20-min session experiment, rather than an intervention program intended to bring long-term changes. Second, it is unlikely that the scripts in the VR meditation were empirically tested by the publishers to support the clinical benefits, although it seems to us that the scripts are similar to those included in evidence-based intervention programs. The foundations and elements of the practices embedded in the Guided Meditation VR program should be examined within the frameworks of existing evidence-based mindfulness-based programs, such as mindfulness-based stress reduction. Further research can seek to determine whether such program-based practices translate well into the VR environment and what the impact of their effectiveness is within it. Third, the participants in the VR condition were not clearly more motivated to continue meditating. Finally, it is unclear whether VR being perceived as more fun and less boring by the participants is actually beneficial. One of the core elements of meditation practice is that the meditators become less driven by a desire for excitement through repeated practices. If VR technology was used as a way to add excitement to contemplative practices, we cannot be certain whether the VR meditation is indeed increasing the motivation to engage in meditation or simply strengthening the habit of desiring external stimulation. Future research should explore these questions.

The present study did not directly measure the level of presence, defined as the awareness of the open mind to the arising experience (Parker et al., 2015). Presence is often associated with the benefits provided by the use of VR technology. We do not know the degree to which participants felt the virtual reality was vivid, immersive, and extensive. Due to this limitation, we are unable to conclude whether the positive changes brought by VR mindfulness meditation was due to the immersive and engaging nature of the VR, or due to other factors such as a novelty effect (Huang, 2020; Koch et al., 2018), defined as the interest stemmed from fascination with the novel technology. Future studies could address this issue by taking subjective, behavioral, and/or physiological measures of presence. We also do not know whether repeated exposure might have a different motivational effect than the single exposure tested here.

The present study serves as a replication and extension of research on the use of VR technology to promote and enhance mindfulness practice among the general population using a more rigorous, more statistically powered experimental design. The study found that VR-assisted mindfulness meditation was useful in increasing state mindfulness, bringing positive changes to state affect, and eliciting positive attitudes toward the mindfulness practice, but these benefits were not particularly stronger than video or audio versions of the same practices. Unfortunately, none of the conditions seemed particularly good at providing motivation to meditate more in the future.

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Data Availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethics Statement The research study was approved by the Institutional Review Board of the Iowa State University.

Informed Consent Statement All participants provided a written informed consent before they participated in the study.

Conflict of Interest The authors declare that they have no conflicts of interest.

References

Ajzen, I. (1991). The theory of planned behavior. *Organizational Behavior and Human Decision Processes, 50*(2), 179–211. https://doi.org/10.1016/0749-5978(91)90020-T

Ajzen, I., & Driver, B. L. (1991). Prediction of leisure participation from behavioral, normative, and control beliefs: An application of the theory of planned behavior. *Leisure Sciences, 13*(3), 185–204. https://doi.org/10.1080/01490409109513137

Ajzen, I., & Manstead, A. S. R. (2016). Changing health-related behaviours: An approach based on the theory of planned behaviour. In M. Hewstone & H. Schut (Eds.), *The Scope of Social Psychology: Theory and Applications* (pp. 43–63). Routledge.

Anderson, T., Suresh, M., & Farb, N. A. (2019). Meditation benefits and drawbacks: Empirical codebook and implications for teaching. *Journal of Cognitive Enhancement, 3*(2), 207–220. https://doi.org/10.1007/s41465-018-00119-y

Arpaia, P., D'Errico, G., De Paolis, L. T., Moccaldi, N., & Nuccetelli, F. (2021). A narrative review of mindfulness-based interventions using virtual reality. *Mindfulness, 13*(3), 556–571. https://doi.org/10.1007/s12671-021-01783-6

Barton, A. C., Sheen, J., & Byrne, L. K. (2020). Immediate attention enhancement and restoration from interactive and immersive technologies: A scoping review. *Frontiers in Psychology*, *11*, 2050. https://doi.org/10.3389/fpsyg.2020.02050

Bennett, S. M. (2018). *A trip to the beach: Experimental investigation of mood, the body, and presence in virtual reality meditation* [Master's thesis, University of Oregon]. http://hdl.handle.net/1794/23804. Accessed 13 Mar 2022

Botella, C., Serrano, B., Baños, R., & García-Palacios, A. (2015). Virtual reality exposure-based therapy for the treatment of post-traumatic stress disorder: A review of its efficacy, the adequacy of the treatment protocol, and its acceptability. *Neuropsychiatric Disease and Treatment, 11*, 2533. https://doi.org/10.2147/ndt.s89542

Bravo, A. J., Pearson, M. R., Wilson, A. D., & Witkiewitz, K. (2018). When traits match states: Examining the associations between self-report trait and state mindfulness following a state mindfulness induction. *Mindfulness, 9*(1), 199–211. https://doi.org/10.1007/s12671-017-0763-5

Cebolla, A., Herrero, R., Ventura, S., Miragall, M., Bellosta-Batalla, M., Llorens, R., & Baños, R. M. (2019). Putting oneself in the body of others: A pilot study on the efficacy of an embodied virtual reality system to generate self-compassion. *Frontiers in Psychology*, *10*, 1521. https://doi.org/10.3389/fpsyg.2019.01521

Chandrasiri, A., Collett, J., Fassbender, E., & De Foe, A. (2019). A virtual reality approach to mindfulness skills training. *Virtual Reality, 24*(1), 143–149. https://doi.org/10.1007/s10055-019-00380-2

Chavez, L. J., Kelleher, K., Slesnick, N., Holowacz, E., Luthy, E., Moore, L., & Ford, J. (2020). Virtual reality meditation among youth experiencing homelessness: Pilot randomized controlled trial of feasibility. *JMIR Mental Health, 7*(9), e18244. https://doi.org/10.2196/18244

Cooper, N., Milella, F., Pinto, C., Cant, I., White, M., & Meyer, G. (2018). The effects of substitute multisensory feedback on task performance and the sense of presence in a virtual reality environment. *PLoS ONE, 13*(2), e0191846. https://doi.org/10.1371/journal.pone.0191846

Costa, M. R., Bergen-Cico, D., Grant, T., Herrero, R., Navarro, J., Razza, R., & Wang, Q. (2019). Nature inspired scenes for guided mindfulness training: Presence, perceived restorativeness and meditation depth. In D. Schmorrow & C. Fidopiastis (Eds.), *Augmented Cognition. HCII 2019. Lecture Notes in Computer Science* (vol. 11580, pp. 517–532). Springer. https://doi.org/10.1007/978-3-030-22419-6_37

Crosswell, L., & Yun, G. W. (2020). Examining virtual meditation as a stress management strategy on college campuses through longitudinal, quasi-experimental research. *Behaviour & Information Technology, 41*(4), 1–15. https://doi.org/10.1080/0144929x.2020.1838609

Davidson, R. J., & Kaszniak, A. W. (2015). Conceptual and methodological issues in research on mindfulness and meditation. *American Psychologist, 70*(7), 581–592. https://doi.org/10.1037/a0039512

De Gauquier, L., Brengman, M., Willems, K., & Van Kerrebroeck, H. (2019). Leveraging advertising to a higher dimension: Experimental research on the impact of virtual reality on brand personality impressions. *Virtual Reality*, *23,* 235–253. https://doi.org/10.1007/s10055-018-0344-5

Delehanty, H. (2017, December 17). The science of meditation. *Mindful*. Retrieved from https://www.mindful.org/meditators-under-the-microscope/. Accessed 13 Mar 2022

Edenfield, T. M., & Saeed, S. A. (2012). An update on mindfulness meditation as a self-help treatment for anxiety and depression. *Psychology Research and Behavior Management, 5*, 131–141. https://doi.org/10.2147/prbm.s34937

Faul, F., Erdfelder, E., Lang, A. G., & Buchner, A. (2007). G*Power 3: A flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behavior Research Methods, 39*(2), 175–191. https://doi.org/10.3758/bf03193146

Flores, A., Linehan, M. M., Todd, S. R., & Hoffman, H. G. (2018). The use of virtual reality to facilitate mindfulness skills training in dialectical behavioral therapy for spinal cord injury: A case study. *Frontiers in Psychology*, *9*, 531. https://doi.org/10.3389/fpsyg.2018.00531

Fuchs, M. (2022, May 17). Virtual reality meditation: Awe, avatars, and psychedelic design. *Washington Post.* Retrieved from https://www.washingtonpost.com/wellness/2022/05/17/virtual-reality-meditation-apps/. Accessed 13 Mar 2022

Ge, Y., & Han, T. (2021). MindJourney: Employing gamification to support mindfulness practice. *Advances in usability, user experience, wearable and assistive technology (AHFE 2021)*, *Lecture Notes in Networks and Systems* (Vol. 275). Springer. https://doi.org/10.1007/978-3-030-80091-8_27

Gibbons, F. X., Gerrard, M., Blanton, H., & Russell, D. W. (1998a). Reasoned action and social reaction: Willingness and intention as independent predictors of health risk. *Journal of Personality and Social Psychology, 74*, 1164–1181. https://doi.org/10.1037/0022-3514.74.5.1164

Gibbons, F. X., Gerrard, M., Ouellette, J. A., & Burzette, R. (1998b). Cognitive antecedents to adolescent health risk: Discriminating between behavioral intention and behavioral willingness. *Psychology and Health, 13*, 319–339. https://doi.org/10.1080/08870449808406754

Godin, G., & Kok, G. (1996). The theory of planned behavior: A review of its applications to health-related behaviors. *American*

Journal of Health Promotion, 11(2), 87–98. https://doi.org/10.4278/0890-1171-11.2.87

Goldenhersch, E., Thrul, J., Ungaretti, J., Rosencovich, N., Waitman, C., & Ceberio, M. R. (2020). Virtual reality smartphone-based intervention for smoking cessation: Pilot randomized controlled trial on initial clinical efficacy and adherence. *Journal of Medical Internet Research, 22*(7), e17571. https://doi.org/10.2196/17571

Gómez-Busto, F., & Ortiz, M. (2020). Virtual reality and psychedelics for the treatment of psychiatric disease: A systematic literature review. *Clinical Neuropsychiatry, 17*(6), 365–380. https://doi.org/10.36131/cnfioritieditore20200606

Goyal, M., Singh, S., Sibinga, E., Gould, N., Rowland-Seymour, A., Sharma, R., Berger, Z., Sleicher, D., Maron, D., Shihab, H., Ranasinghe, P., Linn, S., Saha, S., Bass, E., Haythornthwaite, J., & Cramer, H. (2014). Meditation programs for psychological stress and well-being: A systematic review and meta-analysis. *JAMA Internal Medicine, 57*(3), 357–368. https://doi.org/10.1016/j.dza.2014.07.007

Grow, J. C., Collins, S. E., Harrop, E. N., & Marlatt, G. A. (2015). Enactment of home practice following mindfulness-based relapse prevention and its association with substance-use outcomes. *Addictive Behaviors, 40*, 16–20. https://doi.org/10.1016/j.addbeh.2014.07.030

Haisley, K. R., Straw, O. J., Müller, D. T., Antiporda, M. A., Zihni, A. M., Reavis, K. M., Bradley, D. D., & Dunst, C. M. (2020). Feasibility of implementing a virtual reality program as an adjuvant tool for peri-operative pain control: Results of a randomized controlled trial in minimally invasive foregut surgery. *Complementary Therapies in Medicine, 49*, 102356. https://doi.org/10.1016/j.ctim.2020.102356

Hanley, A. W., Dehili, V., Krzanowski, R. N., Barou, D., Lecy, N., & Garland, E. L. (2022). Effects of video-guided groups vs. solitary meditation on mindfulness and social connectivity: A pilot study. *Clinical Social Work Journal, 50*, 316–324. https://doi.org/10.1007/s10615-021-00812-0

Huang, W. (2020). *Investigating the novelty effect in virtual reality on STEM learning*. [Doctoral dissertation, Arizona State University]. https://keep.lib.asu.edu/items/158443. Accessed 13 Mar 2022

Johnson, S., Gur, R. M., David, Z., & Currier, E. (2015). One-session mindfulness meditation: A randomized controlled study of effects on cognition and mood. *Mindfulness, 6*(1), 88–98. https://doi.org/10.1007/s12671-013-0234-6

Kabat-Zinn, J. (2015). Mindfulness. *Mindfulness, 6*, 1481–1483. https://doi.org/10.1007/s12671-015-0456-x

Kaplan-Rakowski, R., Johnson, K. R., & Wojdynski, T. (2021). The impact of virtual reality meditation on college students' exam performance. *Smart Learning Environments*, *8,* 21. https://doi.org/10.1186/s40561-021-00166-7

Keijsers, M., Vega-Corredor, M. C., Tomintz, M., & Hoermann, S. (2021). Virtual reality technology use in cigarette craving and smoking interventions (I "virtually" quit): Systematic review. *Journal of Medical Internet Research, 23*(9), e24307. https://doi.org/10.2196/24307

Koch, M., von Luck, K. V., Schwarzer, J., & Draheim, S. (2018). The novelty effect in large display deployments-Experiences and lesson-learned for evaluating prototypes. *Proceedings of the 16th European Conference on Computer-Supported Cooperative Work,* 1–19. https://doi.org/10.18420/ecscw2018_3

Kolbe, L., Jaywant, A., Gupta, A., Vanderlind, W. M., & Jabbour, G. (2021). Use of virtual reality in the inpatient rehabilitation of COVID-19 patients. *General Hospital Psychiatry, 71*, 76–81. https://doi.org/10.1016/j.genhosppsych.2021.04.008

Lafortune, D., Dion, L., & Renaud, P. (2020). Virtual reality and sex therapy: Future directions for clinical research. *Journal of Sex & Marital Therapy, 46*(1), 1–17. https://doi.org/10.1080/0092623x.2019.1623357

Lau, M. A., Bishop, S. R., Segal, Z. V., Buis, T., Anderson, N. D., Carlson, L., Shapiro, S., Carmody, J., Abbey, S., & Devins, G. (2006). The Toronto mindfulness scale: Development and validation. *Journal of Clinical Psychology, 62*(12), 1445–1467. https://doi.org/10.1002/jclp.20326

Lau, K. W., & Lee, P. Y. (2019). Shopping in virtual reality: A study on consumers' shopping experience in a stereoscopic virtual reality. *Virtual Reality*, *23,* 255–268. https://doi.org/10.1007/s10055-018-0362-3

Lauricella, S. (2013). Mindfulness meditation with undergraduates in face-to-face and digital practice: A formative analysis. *Mindfulness, 5*(6), 682–688. https://doi.org/10.1007/s12671-013-0222-x

Lymeus, F., Lundgren, T., & Hartig, T. (2017). Attentional effort of beginning mindfulness training is offset with practice directed toward images of natural scenery. *Environment and Behavior, 49*(5), 536–559. https://doi.org/10.1177/0013916516657390

Mazgelytė, E., Rekienė, V., Dereškevičiūtė, E., Petrėnas, T., Songailienė, J., Utkus, A., Chomentauskas, G., & Karčiauskaitė, D. (2021). Effects of virtual reality-based relaxation techniques on psychological, physiological, and biochemical stress indicators. *Healthcare, 9*(12), 1729. https://doi.org/10.3390/healthcare9121729

Mestre, D. R., Dagonneau, V., & Mercier, C.-S. (2011). Does virtual reality enhance exercise performance, enjoyment, and dissociation? An exploratory study on a stationary bike apparatus. *Presence: Teleoperators and Virtual Environments, 20*(1), 1–14. https://doi.org/10.1162/pres_a_00031

Miller, M., Mistry, D., Jetly, R., & Frewen, P. (2021). Meditating in virtual reality 2: Phenomenology of vividness, egocentricity and absorption-immersion. *Mindfulness, 12*(5), 1195–1207. https://doi.org/10.1007/s12671-020-01589-y

Mistry, D., Zhu, J., Tremblay, P., Wekerle, C., Lanius, R., Jetly, R., & Frewen, P. (2020). Meditating in virtual reality: Proof-of-concept intervention for posttraumatic stress. *Psychological Trauma: Theory, Research, Practice, and Policy, 12*(8), 847–858. https://doi.org/10.1037/tra0000959

Mrazek, M. D., Smallwood, J., & Schooler, J. W. (2012). Mindfulness and mind-wandering: Finding convergence through opposing constructs. *Emotion, 12*(3), 442–448. https://doi.org/10.1037/a0026678

Mukamal, R. (2017, April 12). *Are virtual reality headsets safe for eyes?* American Academy of Ophthalmology. Retrieved from https://www.aao.org/eye-health/tips-prevention/are-virtual-reality-headsets-safe-eyes. Accessed 13 Mar 2022

Naumburg, C. (2016, June 27). *5 reasons you're having a hard time being mindful*. Mindful. Retrieved from https://www.mindful.org/5-reasons-youre-hard-time-mindful/. Accessed 13 Mar 2022

Navarro-Haro, M. V., López-del-Hoyo, Y., Campos, D., Linehan, M. M., Hoffman, H. G., García-Palacios, A., Modrego-Alarcón, M., Borao, L., & García-Campayo, J. (2017). Meditation experts try virtual reality mindfulness: A pilot study evaluation of the feasibility and acceptability of virtual reality to facilitate mindfulness practice in people attending a mindfulness conference. *PloS ONE, 12*(11), e0187777. https://doi.org/10.1371/journal.pone.0187777

Navarro-Haro, M. V., Modrego-Alarcón, M., Hoffman, H. G., López-Montoyo, A., Navarro-Gil, M., Montero-Marin, J., García-Palacios, A., Borao, L., & García-Campayo, J. (2019). Evaluation of a mindfulness-based intervention with and without virtual reality dialectical behavior therapy® mindfulness skills training for the treatment of generalized anxiety disorder in primary care: A pilot study. *Frontiers in Psychology*, *10*, 55. https://doi.org/10.3389/fpsyg.2019.00055

Ostrom, T. M. (1969). The relationship between the affective, behavioral, and cognitive components of attitude. *Journal of Experimental Social Psychology, 5*(1), 12–30. https://doi.org/10.1016/0022-1031(69)90003-1

Parker, S. C., Nelson, B. W., Epel, E. S., & Siegel, D. J. (2015). The science of presence: A central mediator of the interpersonal benefits of mindfulness. In K. W. Brown, J. D. Creswell, & R. M. Ryan (Eds.), *Handbook of mindfulness: Theory, research, and practice* (pp. 225–244). The Guilford Press.

Qian, J., McDonough, D. J., & Gao, Z. (2020). The effectiveness of virtual reality exercise on individual's physiological, psychological and rehabilitative outcomes: A systematic review. *International Journal of Environmental Research and Public Health, 17*(11), 4133. https://doi.org/10.3390/ijerph17114133

Riva, G., Waterworth, J. A., Waterworth, E. L., & Mantovani, F. (2011). From intention to action: The role of presence. *New Ideas in Psychology, 29*(1), 24–37. https://doi.org/10.1016/j.newideapsych.2009.11.002

Roquet, C. D., & Sas, C. (2018). Evaluating mindfulness meditation apps. In *CHI '23: Proceedings of the 2023 CHI conference on human factors in computing systems* (pp. 1–6). https://doi.org/10.1145/3170427.3188616

Saks, E. R., Jeste, D. V., Granholm, E., Palmer, B. W., & Schneiderman, L. (2002). Ethical issues in psychosocial interventions research involving controls. *Ethics & Behavior, 12*(1), 87–101. https://doi.org/10.1207/S15327019EB1201_6

Seabrook, E., Kelly, R., Foley, F., Theiler, S., Thomas, N., Wadley, G., & Nedeljkovic, M. (2020). Understanding how virtual reality can support mindfulness practice: Mixed methods study. *Journal of Medical Internet Research, 22*(3), e16106. https://doi.org/10.2196/16106

Sevilla-Llewellyn-Jones, J., Santesteban-Echarri, O., Pryor, I., McGorry, P., & Alvarez-Jimenez, M. (2018). Web-Based mindfulness interventions for mental health treatment: Systematic review and meta-analysis. *JMIR Mental Health, 5*(3), e10278. https://doi.org/10.2196/10278

Slater, M. (2009). Place illusion and plausibility can lead to realistic behaviour in immersive virtual environments. *Philosophical Transactions of the Royal Society b: Biological Sciences, 364*(1535), 3549–3557. https://doi.org/10.1098/rstb.2009.0138

Sliwinski, J., Katsikitis, M., & Jones, C. (2017). A review of interactive technologies as support tools for the cultivation of mindfulness. *Mindfulness, 8*(5), 1150–1159. https://doi.org/10.1007/s12671-017-0698-x

Soh, D. J. H., Ong, C. H., Fan, Q., Seah, D. J. L., Henderson, S. L., Jeevanandam, L., & Doshi, K. (2021). Exploring the use of virtual reality for the delivery and practice of stress-management exercises. *Frontiers in Psychology*, *12*, 640341. https://doi.org/10.3389/fpsyg.2021.640341

Statista. (2023). *U.S. VR and AR users 2017–2022*. Statista.com. Retrieved from https://www.statista.com/statistics/1017008/united-states-vr-ar-users/. Accessed 13 Mar 2022

Tarrant, J., Viczko, J., & Cope, H. (2018). Virtual reality for anxiety reduction demonstrated by quantitative EEG: A pilot study. *Frontiers in Psychology*, *9*, 1280. https://doi.org/10.3389/fpsyg.2018.01280

Tarrant, J., Jackson, R., & Viczko, J. (2022). A feasibility test of a brief mobile virtual reality meditation for frontline healthcare workers in a hospital setting. *Frontiers in Virtual Reality, 3*, 764745. https://doi.org/10.3389/frvir.2022.764745

Waller, M., Mistry, D., Jetly, R., & Frewen, P. (2021). Meditating in virtual reality 3: 360° video of perceptual presence of instructor. *Mindfulness, 12*(6), 1424–1437. https://doi.org/10.1007/s12671-021-01612-w

Weitzman, R. E., Wong, K., Worrall, D. M., Park, C., McKee, S., Tufts, R. E., Teng, M. S., & Iloreta, A. M. (2021). Incorporating virtual reality to improve otolaryngology resident wellness: One institution's experience. *The Laryngoscope, 131*(9), 1972–1976. https://doi.org/10.1002/lary.29529

Wren, A. A., Neiman, N., Caruso, T. J., Rodriguez, S., Taylor, K., Madill, M., Rives, H., & Nguyen, L. (2021). Mindfulness-Based virtual reality intervention for children and young adults with inflammatory bowel disease: A pilot feasibility and acceptability study. *Children, 8*(5), 368. https://doi.org/10.3390/children8050368

Yildirim, C., & O'Grady, T. (2020). The efficacy of a virtual reality-based mindfulness intervention. In *2020 IEEE International Conference on Artificial Intelligence and Virtual Reality (AIVR), Utrecht, Netherlands* (pp. 158–165). https://doi.org/10.1109/AIVR50618.2020.00035

Zhang, S., Chen, M., Yang, N., Lu, S., & Ni, S. (2021). Effectiveness of VR based mindfulness on psychological and physiological health: A systematic review. *Current Psychology, 42*, 5033–5045. https://doi.org/10.1007/s12144-021-01777-6

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